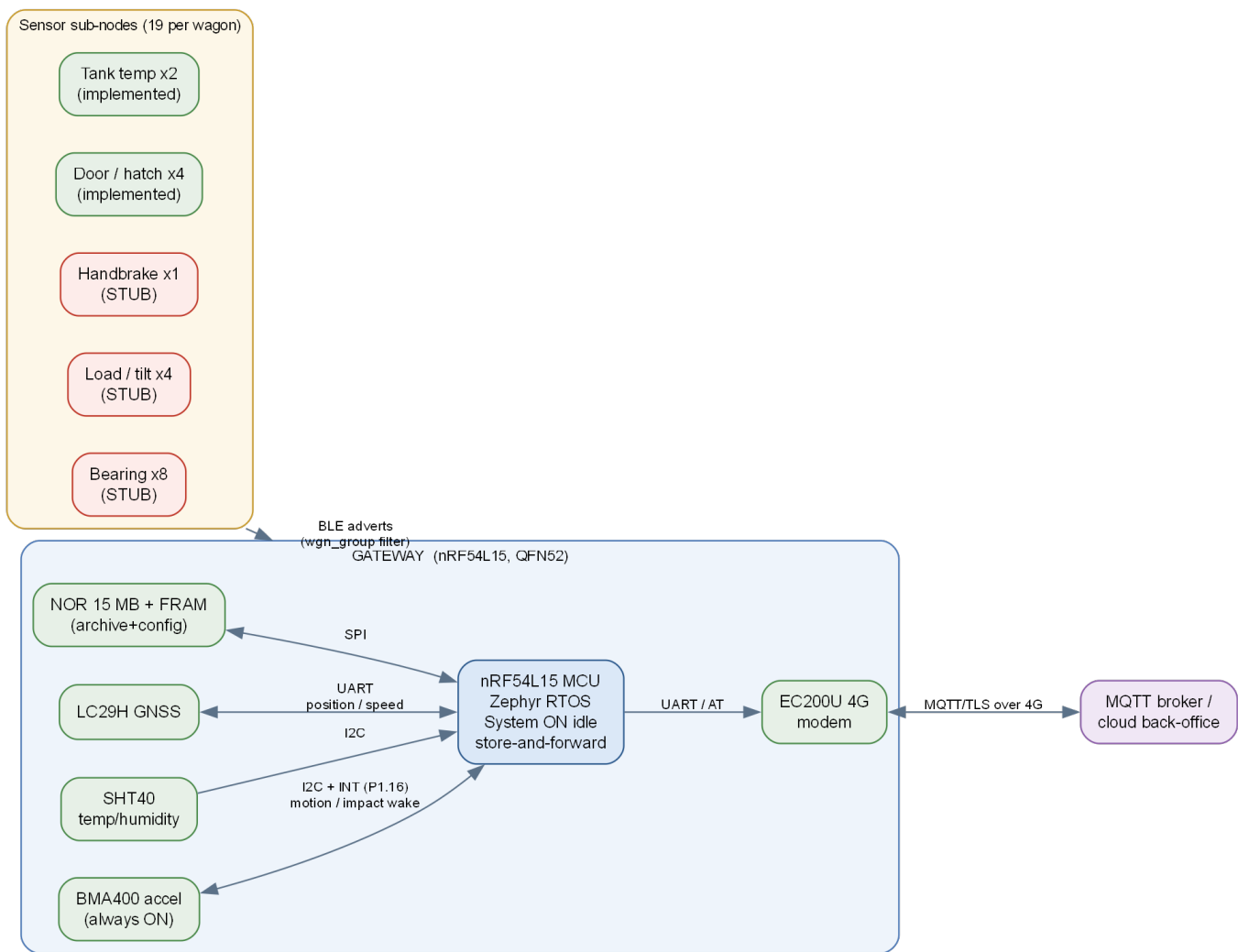


Smart Wagon — System Overview & Sensor Roles

How the gateway, sub-nodes and each sensor fit together · nRF54L15 (QFN52)



1. What the system is

Each freight wagon carries one **gateway** (the nRF54L15 board) and **19 sensor sub-nodes**. The sub-nodes are battery BLE broadcasters that each watch one thing and shout a short encrypted advert. The gateway listens for its own wagon's nodes, adds its **own** measurements (position, motion, climate), and forwards everything to the cloud over 4G - buffering to flash whenever the network is down.

2. Implementation status by node type

Every sub-node board carries a **BMA400 accelerometer** and an **STS4x internal temperature sensor**. Only the DOOR and TANK firmware actually reads them. The other 13 nodes build, encrypt and transmit correctly but report a **fabricated constant**.

Node type	Qty	Sensor driver	What it reports today
DOOR	4	BMA400 + STS4x + reed, implemented	Real internal temp, real peak a , real door state
TANK_TEMP	2	RTD + BMA400 + STS4x, implemented	Real cargo temp (needs calibration) + real internal temp
BEARING	8	none - stub	Fixed 25.0 °C, vibration 0

LOAD_TILT	4	none - stub	Fixed 25.0 °C, secondary 0
HANDBRAKE	1	none - stub	Fixed 25.0 °C, secondary 0

The stub nodes also declare no I2C bus in their devicetree overlay, so the fitted BMA400 and STS4x are not merely unread - they are unreachable until the overlay gains the bus, the rail switch and the interrupt line that the DOOR overlay already has.

3. Why there are two "movement" sensors

BMA400 accelerometer	LC29H GNSS
Question: "Is the wagon vibrating / has it been shocked right now ?" Answers:	"Where is the wagon and how fast is it going?"
Power: microamps - can stay on 24/7	tens of milliamps - short bursts only
Always-on? Yes - it is the wake-up trigger	No - powered only around a fix
Role: Gatekeeper: wakes the CPU, feeds the debounce	Measurer: exact position + speed for the report

The accelerometer notices movement for microamps and wakes the system; only then does the expensive GNSS turn on. Using GNSS to continuously watch for movement would flatten the battery in weeks.

4. Role of every block

Block	Its job
BMA400	Always-on gatekeeper: vibration, high-g impact/derailment, and the CPU wake interrupt (P1.16). On DOOR nodes it also provides the §7.3 always-on tamper watch.
LC29H GNSS	Position and speed. Speed gates bearing-vibration validity (> 15 km/h) and door/handbrake context.
SHT40 (gateway) / STS4x (nodes)	Local climate / node internal temperature. Stateless single-shot reads.
NOR + FRAM	Store-and-forward: ~106 days of reports when 4G is down, replayed with original seq/ts. FRAM holds config and ring pointers.
EC200U 4G	The uplink. Fully powered down between reports.
BLE observer	Caches this wagon's node adverts (filtered by wgn_group) and catches alarm adverts in a 100 ms / 5 s receive window.

5. One report, end to end

1. **Asleep** (System ON idle): only the BMA400 and the BLE observer are awake.
2. **Wake**: schedule timer, BMA400 interrupt, or a node alarm.
3. **Measure**: GNSS warm fix; SHT40 + BMA400; each provisioned node's cached reading (down → SENSOR_FAULT).
4. **Send**: modem on, MQTT connect, publish, drain downlink commands.
5. **Sleep**: GNSS and modem off, re-check debounced motion to set the next cadence, back to idle.